DATABASE RELATIONSHIP ANALYSIS USING JOINS
PROJECT TITLE:SHOPWITHNYKAA
DBMS-WEEK 8

Objectives:

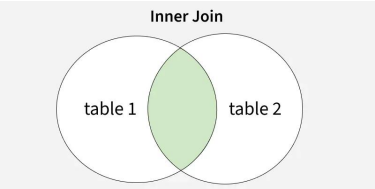
1. Combine Customer, Product, Orders, and Payment tables.
2. Implement INNER JOIN, LEFT JOIN, and RIGHT JOIN.
3. Retrieve complete order details.
4. Display customer purchase history.
5. Generate multi-table reports.

1. Combining Multiple Tables

In your EcoCart project, combining multiple tables means retrieving related information from two or more tables using **JOIN operations**

2. INNER JOIN

INNER JOIN in SQL is used to combine rows from two or more tables based on a matching condition. It returns **only the rows that have matching values in both tables**.



2.1 Customer and Orders

```
SQL> SELECT
  2       c.Customer_ID,
  3       c.Customer_Name,
  4       o.Order_ID,
  5       o.Order_Date,
  6       o.Total_Amount
  7  FROM Customer c
  8  INNER JOIN Orders o
  9       ON c.Customer_ID = o.Customer_ID;

CUSTOMER_ID CUSTOMER_NAME                                          ORDER_ID
----------- ---------------------------------------------------- ----------
ORDER_DAT TOTAL_AMOUNT
--------- ------------
        101 Anitha Kumar                                              201
16-SEP-26          3500

        102 Rahul                                                     202
15-SEP-26          1800

        103 Priya                                                     203
12-SEP-26          3200


CUSTOMER_ID CUSTOMER_NAME                                          ORDER_ID
----------- ---------------------------------------------------- ----------
ORDER_DAT TOTAL_AMOUNT
--------- ------------
        101 Anitha Kumar                                              204
20-SEP-26          1000
```

This displays customers who have placed orders.

2.2 Orders and Payment

```
SQL> SELECT o.Order_ID,
  2         o.Order_Date,
  3         o.Total_Amount,
  4         p.Payment_Mode,
  5         p.Payment_Amount,
  6         p.Payment_Status
  7  FROM Orders o
  8  INNER JOIN Payment p
  9  ON o.Order_ID = p.Order_ID;

  ORDER_ID ORDER_DAT TOTAL_AMOUNT PAYMENT_MODE                    PAYMENT_AMOUNT
---------- --------- ------------ ------------------------------ --------------
PAYMENT_STATUS
------------------------------
       201 16-SEP-26         3500 UPI                                      3500
Successful

       202 15-SEP-26         1800 Card                                     1800
Successful

       203 12-SEP-26         3200 Net Banking                              3200
Successful


  ORDER_ID ORDER_DAT TOTAL_AMOUNT PAYMENT_MODE                    PAYMENT_AMOUNT
---------- --------- ------------ ------------------------------ --------------
PAYMENT_STATUS
------------------------------
       204 20-SEP-26         1000 UPI                                      1000
Successful
```

This displays order and corresponding payment details

2.3 Product and Order_Details

```
SQL> SELECT p.Product_ID,
  2  p.Product_Name,
  3  od.Order_ID,
  4  od.Quantity,
  5  od.Unit_Price,
  6  od.Subtotal
  7  FROM Product p
  8  INNER JOIN Order_Details od
  9  ON p.Product_ID = od.Product_ID;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  ORDER_ID   QUANTITY UNIT_PRICE   SUBTOTAL
---------- ---------- ---------- ----------
         1
Organic Turmeric Powder
       201          3       1000       3000

         2
Black Pepper
       201          1        500        500

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  ORDER_ID   QUANTITY UNIT_PRICE   SUBTOTAL
---------- ---------- ---------- ----------

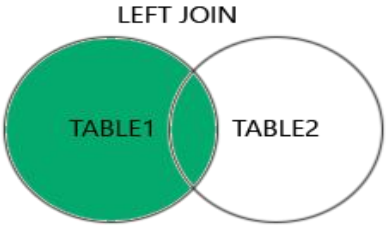
         2
Black Pepper
       202          2        500       1000

         3
Bamboo Toothbrush
```

This displays products that have been included in orders

3. LEFT JOIN

LEFT JOIN returns all records from the left table and matching records from the right table.



3.1 Display All Customers and Their Orders

```
SQL> SELECT c.Customer_ID,
  2  c.Customer_Name,
  3  o.Order_ID,
  4  o.Order_Date,
  5  o.Total_Amount
  6  FROM Customer c
  7  LEFT JOIN Orders o
  8  ON c.Customer_ID = o.Customer_ID;

CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT TOTAL_AMOUNT
--------- ------------
        101 Anitha Kumar                                            201
16-SEP-26         3500

        102 Rahul                                                   202
15-SEP-26         1800

        103 Priya                                                   203
12-SEP-26         3200

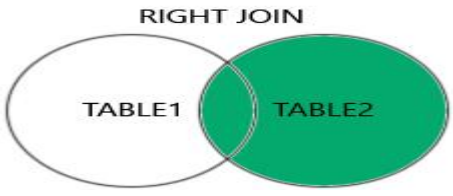

CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT TOTAL_AMOUNT
--------- ------------
        101 Anitha Kumar                                            204
20-SEP-26         1000

        104 Kavin
```

This displays all customers, including customers who have not placed an order.

4. RIGHT JOIN

RIGHT JOIN returns all records from the right table and matching records from the left table.



4.1 Display All Orders and Customer Details

```
SQL> SELECT c.Customer_ID,
  2  c.Customer_Name,
  3  o.Order_ID,
  4  o.Order_Date,
  5  o.Total_Amount
  6  FROM Customer c
  7  RIGHT JOIN Orders o
  8  ON c.Customer_ID = o.Customer_ID;

CUSTOMER_ID CUSTOMER_NAME                                          ORDER_ID
----------- ---------------------------------------------------- ----------
ORDER_DAT TOTAL_AMOUNT
--------- ------------
        101 Anitha Kumar                                              201
16-SEP-26         3500

        101 Anitha Kumar                                              204
20-SEP-26         1000

        102 Rahul                                                     202
15-SEP-26         1800


CUSTOMER_ID CUSTOMER_NAME                                          ORDER_ID
----------- ---------------------------------------------------- ----------
ORDER_DAT TOTAL_AMOUNT
--------- ------------
        103 Priya                                                     203
12-SEP-26         3200
```

This displays all orders along with the matching customer information

5. Complete Order Details

This query combines Customer, Orders, Order_Details, Product, and Payment tables.

```
SQL> SELECT c.Customer_ID,
  2  c.Customer_Name,
  3  o.Order_ID,
  4  o.Order_Date,
  5  p.Product_ID,
  6  p.Product_Name,
  7  od.Quantity,
  8  od.Unit_Price,
  9  od.Subtotal,
 10  py.Payment_Mode,
 11  py.Payment_Amount,
 12  py.Payment_Status
 13  FROM Customer c
 14  INNER JOIN Orders o
 15  ON c.Customer_ID = o.Customer_ID
 16  INNER JOIN Order_Details od
 17  ON o.Order_ID = od.Order_ID
 18  INNER JOIN Product p
 19  ON od.Product_ID = p.Product_ID
 20  INNER JOIN Payment py
 21  ON o.Order_ID = py.Order_ID
 22  ORDER BY o.Order_Date DESC;

CUSTOMER_ID CUSTOMER_NAME                                          ORDER_ID
----------- ---------------------------------------------------- ----------
ORDER_DAT PRODUCT_ID
--------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY UNIT_PRICE   SUBTOTAL PAYMENT_MODE                    PAYMENT_AMOUNT
---------- ---------- ---------- ------------------------------ --------------
PAYMENT_STATUS
------------------------------
        101 Anitha Kumar                                              204
20-SEP-26         3
Bamboo Toothbrush
```

6. Customer Purchase History

This query displays the products purchased by each customer

```
SQL> SELECT c.Customer_ID,
  2  c.Customer_Name,
  3  o.Order_ID,
  4  o.Order_Date,
  5  p.Product_Name,
  6  od.Quantity,
  7  od.Subtotal
  8  FROM Customer c
  9  INNER JOIN Orders o
 10  ON c.Customer_ID = o.Customer_ID
 11  INNER JOIN Order_Details od
 12  ON o.Order_ID = od.Order_ID
 13  INNER JOIN Product p
 14  ON od.Product_ID = p.Product_ID
 15  ORDER BY c.Customer_ID, o.Order_Date;

CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT
---------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY   SUBTOTAL
---------- ----------
         101 Anitha Kumar                                          201
16-SEP-26
Organic Turmeric Powder
         3       3000


CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT
---------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY   SUBTOTAL
---------- ----------
         101 Anitha Kumar                                          201
```

7. Customer Purchase History with Payment

```
SQL> SELECT c.Customer_ID,
  2  c.Customer_Name,
  3  o.Order_ID,
  4  o.Order_Date,
  5  p.Product_Name,
  6  od.Quantity,
  7  od.Subtotal,
  8  py.Payment_Mode,
  9  py.Payment_Status
 10  FROM Customer c
 11  INNER JOIN Orders o
 12  ON c.Customer_ID = o.Customer_ID
 13  INNER JOIN Order_Details od
 14  ON o.Order_ID = od.Order_ID
 15  INNER JOIN Product p
 16  ON od.Product_ID = p.Product_ID
 17  INNER JOIN Payment py
 18  ON o.Order_ID = py.Order_ID
 19  ORDER BY c.Customer_ID;

CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT
---------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY   SUBTOTAL PAYMENT_MODE
---------- ---------- ----------
PAYMENT_STATUS
------------------------------
         101 Anitha Kumar                                          201
16-SEP-26
Organic Turmeric Powder
```

8. Multi-Table Business Report

The following query generates a complete customer purchase report

```
SQL> SELECT c.Customer_ID,
  2  c.Customer_Name,
  3  COUNT(DISTINCT o.Order_ID) AS Total_Orders,
  4  SUM(od.Subtotal) AS Total_Purchase,
  5  SUM(od.Quantity) AS Total_Products
  6  FROM Customer c
  7  INNER JOIN Orders o
  8  ON c.Customer_ID = o.Customer_ID
  9  INNER JOIN Order_Details od
 10  ON o.Order_ID = od.Order_ID
 11  GROUP BY c.Customer_ID, c.Customer_Name
 12  ORDER BY Total_Purchase DESC;

CUSTOMER_ID CUSTOMER_NAME                                    TOTAL_ORDERS
----------- ---------------------------------------------- ------------
TOTAL_PURCHASE TOTAL_PRODUCTS
-------------- --------------
        101 Anitha Kumar                                              2
          4500              9

        103 Priya                                                     1
          3200              4

        102 Rahul                                                     1
          1800              6
```

9. Payment Transaction Report

```
SQL> SELECT py.Payment_ID,
  2  c.Customer_Name,
  3  o.Order_ID,
  4  py.Payment_Date,
  5  py.Payment_Mode,
  6  py.Payment_Amount,
  7  py.Payment_Status
  8  FROM Payment py
  9  INNER JOIN Orders o
 10  ON py.Order_ID = o.Order_ID
 11  INNER JOIN Customer c
 12  ON o.Customer_ID = c.Customer_ID
 13  ORDER BY py.Payment_Date DESC;

PAYMENT_ID CUSTOMER_NAME                                    ORDER_ID
---------- ---------------------------------------------- ----------
PAYMENT_D PAYMENT_MODE                   PAYMENT_AMOUNT
--------- ------------------------------ --------------
PAYMENT_STATUS
------------------------------
       504 Anitha Kumar                                           204
20-SEP-26 UPI                                  1000
Successful

       501 Anitha Kumar                                           201
16-SEP-26 UPI                                  3500
Successful

PAYMENT_ID CUSTOMER_NAME                                    ORDER_ID
---------- ---------------------------------------------- ----------
PAYMENT_D PAYMENT_MODE                   PAYMENT_AMOUNT
--------- ------------------------------ --------------
PAYMENT_STATUS
------------------------------

       502 Rahul                                                  202
15-SEP-26 Card                                 1800
Successful

       503 Priya                                                  203
12-SEP-26 Net Banking                          3200
```

CONCLUSION

In conclusion, this experiment helped us understand database joins and their uses. We used INNER JOIN, LEFT JOIN, and RIGHT JOIN to combine data from multiple tables. We retrieved customer details, order history, product information, and payment reports. Thus, we learned how to retrieve useful information from multiple tables using SQL joins.